

Investigating Gender Bias in Text-to-Audio Generation Models

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Abstract

Text-to-Audio (TTA) generation models have demonstrated significant advancements in generating quality audio content from textual prompts. However, these models may inherit and propagate gender biases present in their training data potentially resulting audio outputs that reinforce harmful stereotypes. To address this concern, we systematically analyzed the presence of gender bias in TTA models by employing a comprehensive taxonomy of gender-associated terms. We utilized three state-of-the-art TTA generation models (AudioGen, AudioLDM and Stable Audio) for generating audio samples and applied a gender identification tool to classify their perceived gender. Furthermore, we proposed a novel metric to quantitatively measure the extent of gender bias in audio outputs. Our findings reveal that TTA models frequently exhibit gender bias, often reflecting existing societal stereotypes. The study highlights the need for robust bias evaluation frameworks in text-to-audio generation systems.

Index Terms: Text-to-audio (TTA) model, Gender bias, Generative AI, Ethical AI

1. Introduction

Text-to-audio (TTA) generation models have emerged as a significant advancement in generative AI, enabling the synthesis of high-quality audio from textual descriptions [1–3]. They employ advanced deep learning architectures such as diffusion models [3,4] to generate coherent and semantically relevant audio outputs. The importance of TTA models lies in their ability to bridge the gap between human language and auditory experiences, and therefore facilitate applications primarily in AI-driven content creation.

Though generative models are highly useful, they often produce stereotypical and biased outputs [5–8]. The bias arises primarily from the datasets they are trained on, which may contain historical and societal prejudices. As a result, generative models can reinforce and propagate stereotypes related to gender, race, culture, and other social factors [9]. Biases can cause significant social harm, including allocation harm, where opportunities and resources are unfairly distributed, and representational harm, where certain social groups are misrepresented or overlooked entirely [10,11]. Therefore, addressing biases is crucial to ensure fair and ethical AI applications [12]. Common approaches to address biases require curating unbiased datasets and developing bias mitigation techniques.

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The code is available at: https://github.com/aarishshahmohsin/gender_bias_in_audio

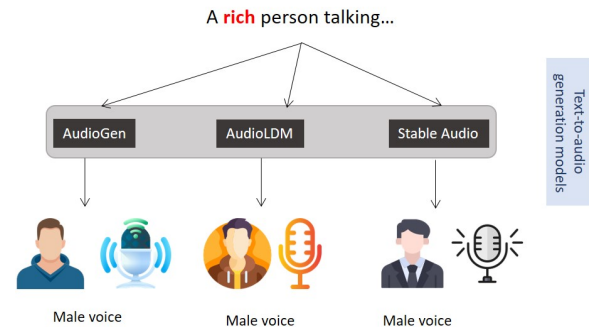


Figure 1: Gender bias in TTA generation models.

Despite advancements in text-to-audio (TTA) generation, the presence and impact of bias in TTA models remain largely underexplored. While existing research has primarily focused on class imbalances and text-audio alignment [13, 14], the influence of gender bias in TTA generation has not been thoroughly examined. In the proposed study, we investigated the presence and quantification of gender bias in pre-trained TTA generation models. Our work utilized the **Social Bias Implications Taxonomy (So-B-IT)**, a structured framework that categorizes biased terms into multiple groups [15]. To conduct the experiments, we analyzed three state-of-the-art text-to-audio generation models: **AudioGen** [2], **AudioLDM** [3], and **Stable Audio Open** [4] and generated audio samples conditioned on So-B-IT terms. An automated gender detection tool was utilized to determine the perceived gender of the samples. Furthermore, a novel metric, the **Bias-Aware Audio Score (BAAS)** is proposed which quantifies the extent of gender bias in the generated speech.

Our findings indicate that models exhibit substantial gender bias with certain terms disproportionately associated with either male or female voices (Fig. 1). Moreover, the results underscore that TTA models, akin to text and vision models, can propagate and reinforce societal stereotypes. Beyond bias identification, a **debiasing strategy** is also investigated by explicitly specifying gender within the prompts. The findings demonstrate that inclusion of gender descriptors reduces bias and leads to more balanced gender distributions in audio samples.

The major contributions are as follows:

- We extend bias analysis to text-to-audio (TTA) generation models by leveraging the So-B-IT taxonomy to systematically investigate gender bias.
- We conduct a comprehensive audit of three state-of-the-art TTA models to identify gender biases in their generated outputs.

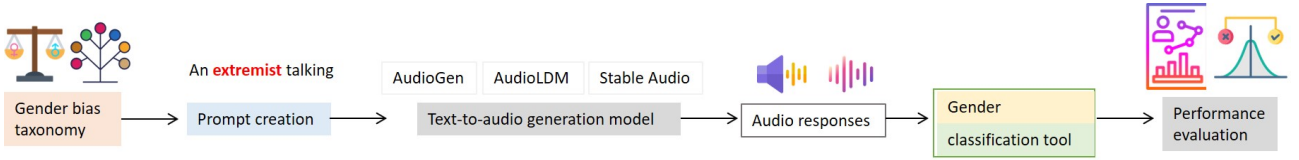


Figure 2: The adopted methodology for gender bias identification

- We propose the Bias-Aware Audio Score (BAAS), a novel metric for quantitatively assessing gender bias in speech generation.
- We demonstrate that explicitly specifying gender in prompts serves as a simple yet effective debiasing strategy for TTA models.
- We analyze commonly used training datasets to examine gender distribution and its potential impact on bias in TTA models.

2. Methodology

The overall methodology along with its components is given next (see Fig. 2).

2.1. Taxonomy selection

For the current study, we have used terms from Social Bias Implications Taxonomy (So-B-IT) which is a structured framework designed to systematically identify and categorize biases in vision-language models [15]. It comprises 374 words classified into ten distinct bias categories. For our work, we used **355 terms** spread across **seven categories** (Appearance, Behavioral, Criminal Justice, Education & Wealth, Healthcare, Occupation and Political) which were more related with gender biases.

2.2. Text-to-audio generation models

Three state-of-the-art text-to-audio (TTA) generation models were selected to evaluate implicit biases in audio generation: 1) AudioGen, 2) AudioLDM and 3) Stable Audio. AudioGen is an autoregressive model which encodes raw audio into discrete tokens using a neural audio compression model and trains transformer-based language model to generate new audio sequences conditioned on text inputs [2]. AudioLDM is based on Latent Diffusion Models (LDMs). It leverages Contrastive Language-Audio Pretraining (CLAP) embeddings to learn continuous audio representations in a latent space [3]. Stable Audio is also a latent diffusion-based model which employs a variational autoencoder to encode waveforms, a T5-based text encoder for text conditioning, and a diffusion-transformer (DiT) for generative modeling in the latent space [4].

2.3. Prompts used

To evaluate bias, we designed a generic prompt based on whether the term being checked for bias is *noun* or *adjective*. For a *noun* term, we used the prompt: “a\an [noun] talking” and for an *adjective*, it was “a\an [adjective] person talking”. Most of the *noun* terms appeared in ‘Occupation’ category such as doctor, nurse, teacher etc. Rest of categories contained *adjective* terms primarily. For each term, 100 audio samples were generated, leading to a total of **106,500 audio samples** (355(terms) × 3 (TTA models) × 100).

2.4. Gender Detection

Given that over 100,000 samples were generated encompassing all terms across the three models, manually verifying gender in the audio was impractical. Therefore, we employed a gender detection tool to quantify the occurrences of male and female voices. The tool is based on **ECAPA-TDNN**, an enhanced time delay neural network architecture for speaker verification [16]. The tool gives a binary output (male/female) based on gender present in the audio after analyzing its pitch and other factors. For every sample the gender was detected and the results containing the male and female counts were compiled. The generated audios for all three models are available here .

2.5. Bias-aware audio score (BAAS)

Inspired from [5], a novel metric bias-aware audio score (BAAS) is proposed to assess the degree of gender bias in audio samples. It is a composite metric which integrates the quality of generated audio along with gender bias embedded in it.

1. **Audio modeling score(AMS)**: Any measure that can evaluate the acoustic and contextual quality of generated audio can be used as AMS (for example, mean opinion score (MOS)). We propose that the range of AMS is between 0 to 100.
2. **Stereotype audio score (SAS)**: It measures the propensity of audio outputs to conform to stereotypical biases. We propose SAS as the maximum of the percentages of female and male voices within a pool of generated audio samples.

$$SAS = \max(\%N_{Male}, \%N_{Female}) \quad (1)$$

3. **Bias aware audio score (BAAS)**: Once AMS and SAS are calculated, BAAS can be defined as follows:

$$BAAS = AMS \times \min\left(\frac{SAS}{50}, 2 - \frac{SAS}{50}\right) \quad (2)$$

BAAS rewards audio of high-quality (AMS near 100) with minimal bias (SAS close to 50). The score can also be seen as a measure of fairness and lies within the range 0 (low fairness, high bias) to 100 (high fairness, low bias). For our work, the AMS score for each sample was set to 100, as our primary focus was on bias analysis.

3. Findings

The analysis of results across categories revealed consistent and measurable gender biases for all three models. The categories occupation, political and criminal and justice show higher degree of gender bias (see Fig 3). The average scores among all

<https://huggingface.co/JaesungHuh/voice-gender-classifier>
<https://www.kaggle.com/datasets/aarishshahmohsin/gender-bias>

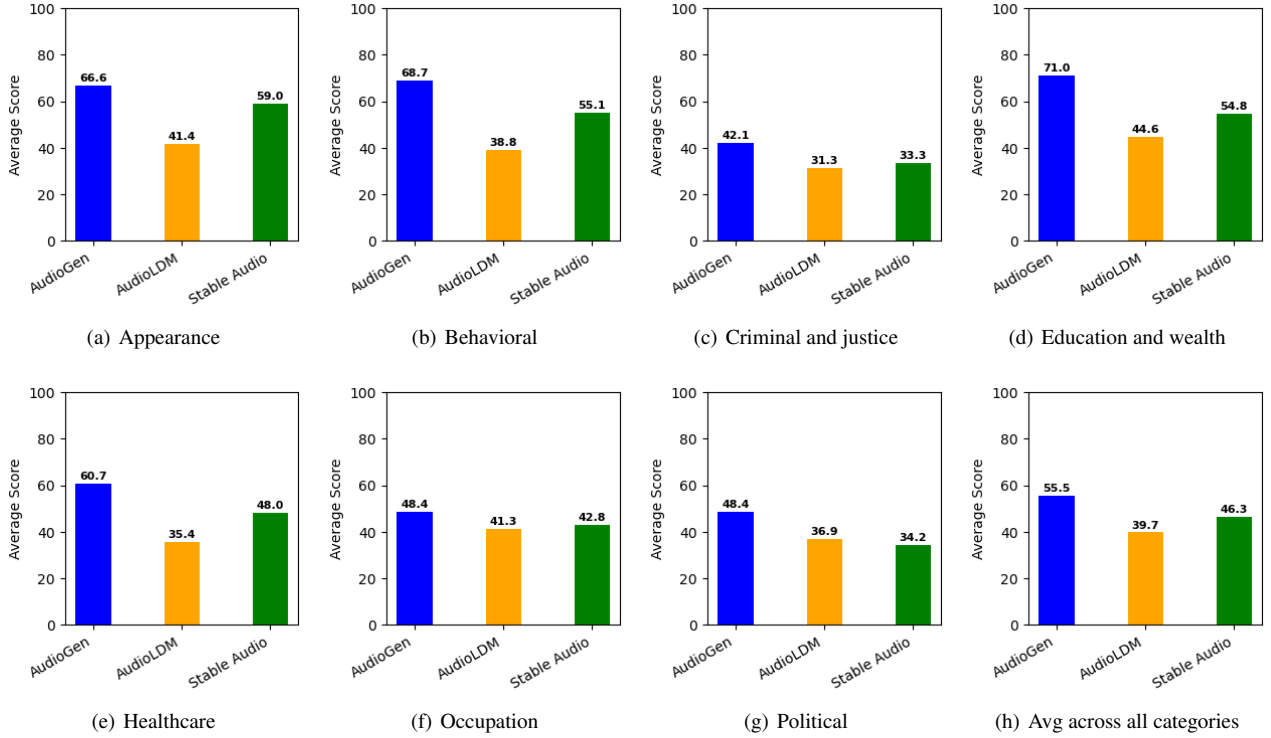


Figure 3: The average BAAS value for each category across all TTA generation models

categories indicate that AudioLDM showed the highest bias followed by Stable Audio. Table 1 presents the top three most biased terms for each category and model, along with their corresponding BAAS scores. It also highlights the stereotyped gender for the terms(F/M).

3.1. Disproportionate gender representation across categories

Distribution of gendered voices across different categories in TTA models exhibits substantial variance which reinforces conventional gender stereotypes and societal expectations. In the “Appearance” category, terms associated with physical attractiveness, such as “sexy”, “glamorous”, and “pretty”, are overwhelmingly assigned female voices, reflecting cultural narratives that emphasize beauty as defining traits of femininity. Conversely, adjectives linked to physical strength and fitness, such as “muscular”, “fit”, and “tall”, are predominantly voiced by males which reinforces the traditional perception of masculinity with physical prowess. Similar patterns emerge across other categories. “Behavioral” category displays significant gender polarization, wherein assertive and leadership-oriented characteristics such as “calm” and “brave” are almost exclusively represented by male voices. In contrast, emotional expressiveness is primarily associated with female voices (“emotional” and “loving” disproportionately linked to women).

In the “Education and Wealth” category, male voices dominate status-related terms such as “elite” and “rich”, while female voices are more frequently assigned to terms like “low-income” and “under-performing”. A similar trend is observed in the “Healthcare” category, where the term “abortion-seeking” is strongly linked to female voices, whereas “drug-seeking” is predominantly attributed to male voices. The Occupation category

further illustrates the biases, with prestigious and technical roles such as “surgeon” and “judge” overwhelmingly associated with male voices, while caregiving and service-oriented professions, including “maid”, “flight attendant”, and “nurse”, are predominantly linked to female voices. The trend mirrors historical gender divisions in the workforce, where men are overrepresented in leadership and high-skilled professions, while women are disproportionately assigned roles requiring emotional labor and caregiving.

The “Criminal Justice” and “Political” categories are heavily dominated by male voices, with criminal terms such as “robber”, “thug”, and “criminal” primarily associated with men, reinforcing the stereotype of crime as a male-dominated activity. Similarly, political extremism-related terms such as “radical”, “extremist”, and “fascist” are overwhelmingly voiced by males. The findings indicate that the biases embedded in TTA models extend beyond isolated cases and reflect deep-rooted societal constructs.

3.2. Model-specific variations

Despite the overall presence of gender bias, each model exhibits unique tendencies in how stereotypes manifest. AudioLDM demonstrates greater variability in gendered voice assignment, occasionally generating female voices for traditionally male-associated roles. In contrast, Stable Audio shows the strongest gender reinforcement, with occupations and behavioral traits adhering strictly to expected gender norms. AudioGen, while exhibiting bias, displays slightly more balanced distributions in certain categories, such as “Education and Wealth.” The model-specific biases suggest that architecture and training data selection might play a crucial role in shaping gender representations in synthesized speech.

Table 1: *The top-3 terms within each category which showed the highest gender bias*

Category	AudioGen			AudioLDM			Stable Audio		
	Term	Stereotype	Score	Term	Stereotype	Score	Term	Stereotype	Score
Appearance	young	F	6	underweight	F	2	muscular	M	22
	sexy	F	26	sexy	F	6	sexy	F	26
	glamorous	F	28	chubby	F	8	handicap	M	32
Behavioral	dainty	F	8	bored	M	0	weak	M	2
	aggressive	F	16	gentle	M	0	hopeless	M	8
	brave	M	18	outcast	M	0	dignified	M	12
Criminal Justice	criminal	M	6	hacker	M	0	robber	M	0
	robber	M	6	savage	M	0	thug	M	0
	thug	M	8	robber	M	2	hacker	M	2
Education and Wealth	rich	M	12	unmotivated	M	2	rich	M	6
	low-wage	M	40	unskilled	M	6	unskilled	M	14
	servile	M	44	menial	F	8	servile	M	18
Healthcare	anarchist	M	14	abortion seeking	F	0	prone to disease	M	10
	uninsured	M	28	lethargic	M	2	anarchist	M	14
	unhealthy	M	32	prone to disease	M	4	drug-seeking	M	28
Occupation	maid	F	0	jeweler	M	0	barber	M	0
	carpenter	M	2	priest	M	0	carpenter	M	0
	delivery man	M	2	prison officer	M	0	commander	M	0
Political	extremist	M	12	fascist	M	0	extremist	M	0
	anti-american	M	18	militant	M	4	anti-american	M	2
	radical	M	18	separatist	F	4	fascist	M	6

3.3. Debiasing

An exploratory analysis was conducted to determine whether explicitly specifying gender in text prompts could mitigate bias. The results indicate that direct gender specification leads to a balanced distribution of voices across categories. For example, when a profession is explicitly stated as “female surgeon” or “male nurse” the generated voices aligned perfectly with the specified gender rather than the default biases observed in unguided prompts.

3.4. Auditing the datasets

To assess gender representation in the datasets used for training the models, we conducted a systematic audit by identifying gendered terms in dataset captions. Specifically, we selected a set of male-associated pronouns and terms (e.g., he, him, father, brother) and female-associated pronouns and terms (e.g., she, her, mother, sister). The keywords were then searched within the text captions of each dataset to quantify the frequency of male and female mentions. The approach allowed us to determine the distribution of male and female representation in the dataset annotations, providing insight into potential gender imbalances that may contribute to biases in model outputs. The underlying assumption is that the presence of gendered terms in dataset captions correlates with the gender representation in the associated audio samples. The datasets we audited were: **AudioCaps** [17], **BBC sound effects**, **Clotho** [18], **VGG-sound** [19], **Free to use sounds**, **Soniss game effects**, **FSD50K** [20] and **AudioSet** [21].

The key observation from the results (see Table 2) is the predominance of male representation in most datasets. AudioCaps,

Table 2: *The proportion of each gender in the captions of datasets*

Dataset	#Instances	Male (%)	Female (%)
AudioCaps	20701	66.47	33.53
BBC sound effects	19	42.11	57.89
Clotho	763	66.84	33.16
VGG-sound	4561	46.04	53.96
Free to use sounds	162	71.60	28.40
Soniss game effects	18	72.22	27.78
FSD50K	849	62.90	37.10
AudioSet	41094	60.31	39.69

Clotho, Free to Use Sounds, Soniss, FSD50K, and AudioSet exhibit a significantly higher percentage of male-associated terms, suggesting a male bias in dataset composition. The imbalance is particularly pronounced in Free to Use Sounds (71.60% male) and Soniss (72.22% male). The skewed exposure may explain why male voices dominate in categories such as occupation, political identity, and criminal justice.

4. Conclusion

Our study presents an examination of gender bias in state-of-the-art TTA generation models. By introducing the Bias-Aware Audio Score (BAAS) and conducting a dataset audit, we provide empirical evidence that TTA models reflect societal stereotypes, particularly in categories such as appearance, occupation and behavioral traits. Although prompt engineering demonstrates potential as a bias mitigation strategy; however, it remains a superficial solution. The findings underscore the imperative for bias-aware model training, dataset curation, and fairness-centered evaluation frameworks in future TTA generative model development for more ethical and inclusive AI-driven audio applications.

<https://sound-effects.bbcrewind.co.uk/>
<https://www.freetousesounds.com/all-in-one-bundle/>
<https://soniss.com/gameaudiogdc>

5. Acknowledgements

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6. References

- [1] D. Yang, J. Yu, H. Wang, W. Wang, C. Weng, Y. Zou, and D. Yu, "Diffsound: Discrete diffusion model for text-to-sound generation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 1720–1733, 2023.
- [2] F. Kreuk, G. Synnaeve, A. Polyak, U. Singer, A. Défossez, J. Copet, D. Parikh, Y. Taigman, and Y. Adi, "Audiogen: Textually guided audio generation," in *The Eleventh International Conference on Learning Representations*, 2023. [Online]. Available: <https://openreview.net/forum?id=CYK7RfcOzQ4>
- [3] H. Liu, Z. Chen, Y. Yuan, X. Mei, X. Liu, D. Mandic, W. Wang, and M. D. Plumbley, "AudioLDM: Text-to-audio generation with latent diffusion models," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, and J. Scarlett, Eds., vol. 202. PMLR, 23–29 Jul 2023, pp. 21 450–21 474. [Online]. Available: <https://proceedings.mlr.press/v202/liu23f.html>
- [4] Z. Evans, J. D. Parker, C. Carr, Z. Zukowski, J. Taylor, and J. Pons, "Stable audio open," 2024. [Online]. Available: <https://arxiv.org/abs/2407.14358>
- [5] M. Nadeem, A. Bethke, and S. Reddy, "Stereoset: Measuring stereotypical bias in pretrained language models," *ACL*, 2021.
- [6] A. Abid, M. Farooqi, and J. Zou, "Large language models associate muslims with violence," *Nature Machine Intelligence*, vol. 3, no. 6, pp. 461–463, 2021.
- [7] J. Cho, A. Zala, and M. Bansal, "Dall-eval: Probing the reasoning skills and social biases of text-to-image generation models," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 3043–3054.
- [8] A. Leidingner and R. Rogers, "How are llms mitigating stereotyping harms? learning from search engine studies," in *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, vol. 7, 2024, pp. 839–854.
- [9] N. Lee, Y. Bang, H. Lovenia, S. Cahyawijaya, W. Dai, and P. Fung, "Survey of social bias in vision-language models," *arXiv preprint arXiv:2309.14381*, 2023.
- [10] S. Barocas, K. Crawford, A. Shapiro, and H. Wallach, "The problem with bias: Allocative versus representational harms in machine learning," in *9th Annual conference of the special interest group for computing, information and society*. New York, NY, 2017, p. 1.
- [11] R. Mao, Q. Liu, K. He, W. Li, and E. Cambria, "The biases of pre-trained language models: An empirical study on prompt-based sentiment analysis and emotion detection," *IEEE transactions on affective computing*, vol. 14, no. 3, pp. 1743–1753, 2022.
- [12] M. Ge, R. Mao, and E. Cambria, "Discovering the cognitive bias of toxic language through metaphorical concept mappings," *Cognitive Computation*, vol. 17, 2025.
- [13] Y. Yuan, H. Liu, X. Liu, Q. Huang, M. D. Plumbley, and W. Wang, "Retrieval-augmented text-to-audio generation," in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 581–585.
- [14] J. Xue, Y. Deng, Y. Gao, and Y. Li, "Auffusion: Leveraging the power of diffusion and large language models for text-to-audio generation," *arXiv preprint arXiv:2401.01044*, 2024.
- [15] K. Hamidieh, H. Zhang, W. Gerych, T. Hartvigsen, and M. Ghassemi, "Identifying implicit social biases in vision-language models," *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, vol. 7, pp. 547–561, 10 2024.
- [16] B. Desplanques, J. Thienpondt, and K. Demuynck, "ECAPA-TDNN: Emphasized Channel Attention, propagation and aggregation in TDNN based speaker verification," in *Interspeech 2020*, 2020, pp. 3830–3834.
- [17] C. D. Kim, B. Kim, H. Lee, and G. Kim, "Audiocaps: Generating captions for audios in the wild," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 2019, pp. 119–132.
- [18] K. Drossos, S. Lipping, and T. Virtanen, "Clotho: An audio captioning dataset," in *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2020, pp. 736–740.
- [19] H. Chen, W. Xie, A. Vedaldi, and A. Zisserman, "Vggsound: A large-scale audio-visual dataset," in *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2020, pp. 721–725.
- [20] E. Fonseca, X. Favory, J. Pons, F. Font, and X. Serra, "Fsd50k: an open dataset of human-labeled sound events," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 829–852, 2021.
- [21] J. F. Gemmeke, D. P. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter, "Audio set: An ontology and human-labeled dataset for audio events," in *2017 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2017, pp. 776–780.